HardRock Field Equipment Service Manual

Reclaimers • Vermeer and Ditch Witch drills • Tulsa Rig Iron TT 660 pumps

Purpose: Provide crews with a consistent inspection and maintenance routine that identifies plugging, lubrication failures, contamination and wear before they cause downtime. Record each condition, correct the underlying cause and verify the equipment is safe and ready before returning it to service.

Use this manual by equipment system. Begin with the safety and crew-responsibility requirements, then go directly to the applicable reclaimer, drilling rig, mud pump or model-specific lubrication section. Equipment-specific manufacturer instructions and approved asset service cards govern whenever they are more restrictive.

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Safety and crew responsibilities

This section establishes the minimum controls that apply before inspection, cleaning, adjustment or repair. Equipment-specific procedures later in this manual do not replace lockout/tagout, pressure isolation, mechanical support or qualified-person requirements.

Before hands on maintenance

- Park securely, unload the mechanism and establish an exclusion area. Lower components to a stable position or install the manufacturer’s rated mechanical supports. Do not rely on hydraulic cylinders to hold a raised load.
- Stop equipment and isolate all energy sources, including remote starts, generators, electrical feeds, hydraulic accumulators, air and gravity. Apply the company lockout/tagout procedure and verify isolation. A stopped engine alone is not isolation.
- Isolate and bleed trapped pressure on both sides of valves, strainers, cones, pumps and hoses. Confirm zero pressure through the approved procedure; a plugged gauge may falsely indicate zero. Do not loosen fittings to test for pressure.

- Wait for hot fluids and surfaces to cool. Never open a hot cooling system. Never check a pressurized leak with a hand. Treat suspected fluid injection as an emergency.
- Restore guards and covers, remove tools, clear personnel and conduct a controlled restart. Recheck the repaired area from a safe position.

Crew boundaries

Role	Responsibility
Operator	Walkaround, readings, mud checks and observation from guarded access. Stop and report abnormal conditions. Perform only authorized tasks.
Crew leader	Assign named rig, reclaimer and pump owners each shift. Verify records and unresolved defects at handover.
Qualified mechanic	Internal pump service, alignment, electrical repairs, relief and dampener servicing, torque-controlled repairs and diagnosis.
Equipment manager	Approve exact fluids, lubricant maps, OEM intervals, limits and replacement parts. Track due hours and approve return to service.

Stop work and notify the crew leader immediately when any of the following conditions are observed:

- Loss of engine or pump lubrication pressure; any gearbox below its specified oil level; overheating; milky oil, metal or significant foaming. Correct low gearbox oil before operating; report unexplained loss or contamination for mechanic evaluation.
- New knocking, severe vibration, smoke, burning smell, damaged guards or structural cracks.
- High-pressure leak, failed relief device, damaged pressure hose, persistent cavitation or loss of piston cooling/wash.
- Unsafe access, failed safety control, unstable anchoring or unexpected machine movement.

Observation while running does not authorize reaching through guards, touching rotating equipment or opening a pressurized component. Tank entry requires the company confined-space evaluation and entry procedure.

Reclaimer systems

Reclaimer startup and operating routine

HardRock proposed inspection frequency: before each shift, after stable startup, at least hourly while processing, after a ground or flow change, and after any plug or repair. Increase checks when solids loading rises. [R1 R2]

Before startup

- Identify each centrifugal pump by duty: transfer, cone feed, mixing or charge pumps. Verify the correct valve lineup and that hoses cannot collapse, kink or pull loose.
- Check engine/generator oil, coolant when cold, hydraulic level where installed and fuel. Inspect leaks, cooling surfaces, belts, love joy connection, guards, emergency controls and walkways. Use the asset card for level-check conditions.
- Inspect screen mesh, seals, wedges and supports. Replace torn screens and correct bypass paths. Check shaker mounts and transport locks in the operating position required by the OEM.
- Inspect tank levels, compartment flow paths, suction intakes, agitator guards and accessible debris. Remove trash with the system isolated; keep tools and loose materials out of tanks.
- Verify each pump is primed and its suction path available. Confirm required seal or packing support systems are working. Start in the OEM sequence.

After stable startup

- Record cone manifold pressure, pump speed setting, tank level and mud condition as the shift baseline. Use only a verified gauge and the approved range for that cone bank; do not assume one pressure fits every reclaimer.
- Observe each numbered cone, screen loading and tank transfer. Mark a cone that has a different discharge pattern. Confirm all installed cones are operating as intended.

At least hourly and after changes

Check	Action
Every cone	Observe underflow from a safe location. Log number and condition; investigate a missing, roping or changed discharge.
Centrifugals	Record pressure trend; observe leaks, vibration, sound and safe instrument temperature readings.
Screens and tanks	Check tears, blinding, bypass, flooding, adequate fluid level and air entering the suction.
Mud quality	Check sand content, density and viscosity at designated sample points against the project mud program. Trend results; do not use a universal sand limit.
Response	Tell driller immediately if clean-fluid supply or quality is compromised. Do not hide a problem by diluting mud or raising speed.

Hydrocyclone checks and safe clearing

Number and inspect every cone

Label each cone and bank permanently so crews can identify repeat plugging. HardRock proposed standard: observe after startup and at least hourly; check again immediately after a formation change, a pressure change or a repaired screen. A weekly inspection alone will miss operating plugs.

Observation	Possible cause and next check
No discharge from one cone	Possible apex/inlet blockage or low solids loading. Compare neighboring cones, bank pressure and feed condition. Absence of solids alone does not prove a plug.
Roping or a solid narrow stream	Possible overload, restricted apex or unsuitable operating condition. Check feed rate, mud properties and OEM setup.
Changed or uneven spray	Inspect for blockage or wear after isolation. Compare with the normal pattern for that cone design.
Most cones change together	Check tank level, manifold gauge, valve lineup, suction restriction, air ingress and centrifugal performance.
Sand rises despite normal pressure	Inspect screen bypass, cone wear and sample points. Pressure alone does not demonstrate separation quality.

Clearing a suspected blocked cone

- Notify the driller and establish a safe fluid-supply plan. Stop and isolate the affected bank and its pump as permitted by the system design.
- Lock out the energy sources, isolate incoming and backflow paths, drain safely and verify all trapped pressure is relieved. Wear required splash protection.
- Remove the apex or designated access section only as specified by the cone manufacturer. Remove the obstruction with approved tools. Never insert a hand, wire or rod into an operating cone.
- Inspect the inlet, apex and accessible cone wall for damage, grooves or erosion. Replace damaged components with the specified size; do not drill the apex larger.
- Reassemble with correct seals and fasteners. Restore guards, open the approved flow path and restart under control. Verify pressure and the discharge of every cone; repeat the mud-quality check.
- Log cone number, obstruction or wear found, parts changed and verified restart condition. Repeated plugs require investigation upstream rather than repeated clearing alone.

Centrifugal pump care

Identify the lubrication and seal arrangement

Record pump make/model, bearing lubrication type and seal arrangement on the asset card. Bearing grease, packing/seal lubrication and mechanical-seal flush are different functions. Do not put grease in an unidentified port. Never apply a generic shot count to all centrifugal pumps.

When	Field procedure
Before each shift	Check oil level if oil lubricated, or service status if grease lubricated. Check seal/packing support supply, guards, intake availability, hoses and casing leaks.
At startup	Prime using the installed pump procedure. Confirm the authorized valve lineup and operating range. Do not run dry or leave a pump operating at zero flow.
At least hourly proposed	Observe pressure, tank level, sound, vibration and leakage from safe access. Compare instrument temperatures with baseline and OEM limits; do not touch a bearing to judge heat.
After a pressure decline	Check fluid properties and gauge condition, then level, suction restriction, air ingress, valves and drive speed. Internal inspection comes after isolation.

When	Field procedure
At the OEM lubrication interval	Use the exact product, measured quantity and procedure for that bearing housing. Record hours, quantity and fitting ID. No grease is added to sealed bearings unless specified.

Seal and packing rules

- Mechanical seal: maintain the specified liquid supply and priming condition. New leakage or rapid heating requires investigation. Do not treat a mechanical seal like an adjustable packing gland.
- Packed pump: apply the manufacturer's leakage, cooling and adjustment procedure. Do not tighten a gland to eliminate all leakage; excessive compression can overheat packing and wear the shaft sleeve. Never adjust near exposed moving parts.

Troubleshooting before replacing parts

- Low cone pressure can come from a plugged suction, low fluid level, air leak, wrong speed, worn impeller or a gauge problem. Verify the simple causes before increasing speed or ordering an impeller. [R2]
- Rattling, erratic output or rapid wear: reduce load and stop if persistent. Investigate suction conditions and available feed. Restart only after correcting the cause.
- Hot bearing or repeated seal failure: check lubricant quantity/type, contamination, alignment, pipe strain and operating point. Internal measurements and alignment belong to a qualified mechanic.

Reclaimer shutdown and periodic care

Every shift shutdown proposed

- Coordinate with the driller before interrupting supply. Follow the approved shutdown sequence.
- Flush designated mud circuits using an approved source and discharge route. Account for additive compatibility and environmental containment.
- Stop and isolate before screen removal or hands-on cleaning. Wash mesh gently; do not use pressure that damages screens. Clean accessible buildup from decks and safe tank access points.
- Drain or winterize trapped-fluid spaces when freezing is possible, including pump casings, cones, hoses and low points. Use the product and method approved for each component. Do not assume drained tanks mean drained pumps.
- Record worn screens, cone problems, pressure trends, pump leakage and parts used. Leave clean-fluid supply and defects clearly identified for the next shift.

Weekly proposed condition review

- With equipment isolated, inspect pump mountings, couplings, belts, guards, accessible hoses, shaker isolation components and structural fasteners. Use OEM torque and belt-tension values; do not tighten indiscriminately.
- Review cone and sand-test logs. Inspect worn or repeatedly troublesome cones internally; check the remaining bank at the OEM interval or earlier when conditions warrant.
- Inspect accessible intakes and compartment buildup; schedule safe cleaning before loss of working volume or restricted feed. Inspect electrical leads and connectors without opening energized panels.

Monthly proposed review and planned service

- Equipment manager reviews operating hours and upcoming engine, generator, hydraulic, pump, bearing and drive services. OEM hour/calendar limits govern oil, filter and lubricant replacement.
- Qualified mechanic inspects impellers, wear surfaces, seals and bearings when pressure trends, leaks or operating hours justify opening the pump. Measure against manufacturer discard limits.

Before transport or extended storage

- Drain/clean as specified, secure hoses and accessories, install transport locks and inspect running gear, brakes, lights and tires. Verify transport weight and support instructions. Preserve pumps and exposed surfaces under the equipment manual.

Stock by asset: screens and seals, correctly sized cone/apex parts, approved centrifugal seal/packing parts, filters and fluids. Confirm part numbers before dispatch.

Drilling rig systems

Rig startup and fluid inspection

Applies to all listed Vermeer and Ditch Witch rigs where the component is installed. Travel and rotary gearbox oil levels must be checked DAILY BEFORE STARTUP under the proposed HardRock standard. OEM checking methods, oil grades and change intervals still apply.

System	What to check and how
Engine oil	Park in the specified level position, stop and allow the required drain-back time. Read dipstick, inspect for unexplained rise or contamination and fill only to the approved range.
Cooling system	Check reservoir and inspect hoses/cooling pack when cold. Use the specified coolant and concentration. Do not mix unknown coolants or open a hot cap.
Hydraulic reservoir	Use the specified attachment positions and oil temperature. Inspect sight glass/dipstick, leaks and breather. Clean fill area; use filtered transfer equipment.
Travel planetary gearboxes	DAILY BEFORE STARTUP: check both left and right travel/final-drive gearboxes, plus any separate travel reduction boxes. Park level, stop and isolate; let oil settle and cool. Position each level plug/sight glass exactly as the OEM specifies. Verify oil at the specified level; top up only with the approved gear oil. Check every separate reservoir, not just one side.
Rotary planetary gearboxes	DAILY BEFORE STARTUP: check each rotary/rotation gearbox, planetary or reduction housing where fitted. Use the OEM checking position and indicator/level port. Confirm the specified oil level and top up with approved gear oil. Hydraulic reservoir level does not confirm gearbox oil level. Record the actual check and any oil added.
Fuel and water separator	Verify clean fuel supply and contamination indication. Drain only by the engine procedure into containment. Replace filters by engine interval/restriction criteria.
DEF where fitted	Use approved clean DEF and dedicated transfer equipment. Check level and contamination; follow engine regeneration instructions.
Air intake and cooling surfaces	Check restriction indication, hoses and clamps. Inspect cooling pack. Clean by the OEM method; do not routinely open intake plumbing or blow out elements unless permitted.
Onboard mud system	Check pump oil/support fluids where installed, hoses, swivel leaks, valves, relief protection and winterization status.
Thread compound supply	Check quantity and delivery each shift. Verify application at every connection using approved drill-pipe compound. [R5]

Walkaround before drilling

- Inspect carriage/track, racks or chains where installed, vise jaws, anchors, stabilizers, pipe loader, guarding, hoses and power cables. Look for wear, cracks, loose hardware and trapped debris with energy isolated.
- Perform safety-control checks in the operator manual from the designated positions with the area clear. Record failures; do not bypass interlocks to keep drilling.

Rig lubrication and operating care

Grease only identified points

Before each shift, review due points on the signed asset map. The locations below are inspection categories, not a verified fitting list for any model. Grease is applied only at the OEM interval and quantity or an approved documented severe-service interval.

Component to locate on the asset	Required instruction on its service card
Carriage rollers, guides and wear interfaces	Point IDs, access position, lubricant or dry-service requirement and quantity. Do not grease an entire rack or rail by assumption.
Thrust drive chains or rack drive	Approved chain lubricant/gear treatment if required; application method and interval. Keep incompatible grease off designated dry surfaces.
Vise and breakout pivots/slides	Fitting locations and allowable grease surfaces. Keep jaws and gripping surfaces free of lubricant unless the OEM directs otherwise.
Pipe loader, lift arms and pipe rack pivots	Every fitted pivot ID, safe supported position, product, amount and interval.
Anchor, stabilizer and mast pivots	Safe position and support, fitting IDs and interval. Do not work beneath unsupported components.
Rotation head, swivel and bearings	Separate grease points from gearbox fill ports and sealed components. Specify seal-compatible product and quantity.
Undercarriage and track adjuster	Use the exact procedure. A track-tension grease fitting is a pressurized adjustment system, not a routine bearing grease point.
Remote grease manifolds	Confirm each line reaches the intended point, inspect damage/blockage and identify individual outlets.

Correct grease method

- Clean the fitting and coupler, use a labeled dedicated gun and apply the specified measured quantity. Calibrate gun output when grams are specified. Do not assume a shot is a standard volume.
- Stop if grease will not enter or the fitting/line leaks. Report the failed point rather than increasing pressure indefinitely. Wipe excess; record point IDs and hours. Never rotate a shaft by hand unless isolated and permitted.

During operation and at shutdown proposed

- Observe fluid temperature and pressure alarms continuously; check guarded leaks, rod-compound application and unusual drive noise at least every two hours and at handover. Stop for new knocking, movement or leaks.
- After shutdown and isolation, remove mud from inspection and lubrication areas by the approved cleaning method. Reinspect jaws, pipe threads, wear components and hoses. Perform due lubrication at the OEM-specified condition.

Mud pump systems

TT 660 startup and operating checks

Confirm Tulsa Rig Iron TT-660 on the nameplate and identify the engine, drive/chainbox, charge pump, piston/liner size, wash system and relief device. Pump speed, pressure and flow limits must match the installed configuration; horsepower or model name alone does not define the operating limit. [R6]

Before each shift proposed

- With the pump stopped and level, check power-end oil and separate chainbox level at the specified indicators. Inspect color, water, metal and leaks. Use the approved asset card; do not treat nominal capacity as a fill instruction.
- Inspect engine oil/coolant/fuel, belts, cooling surfaces, drive guards and mounts. Check the lubrication system and its pressure indication according to the OEM startup procedure.
- Check mud tank level, open and unobstructed suction/feed path, sound suction hose and correct valve lineup. Inspect the charge pump and required inlet-pressure indication.
- Confirm piston/liner wash reservoir is clean and adequately filled, strainer and nozzles are clear, and each cylinder receives the specified wash/cooling supply. Do not run the pump with an ineffective wash system.
- Inspect guarded fluid-end area for leakage, loose parts and damaged rods/liners. Verify correct expendables and the rated operating envelope for the installed piston size.
- Verify pressure gauge condition, rated discharge plumbing and relief-device configuration. Never blank, defeat or isolate required protection. Use the exact TRI shear-pin combination specified for the desired setting; never mix pin sizes. [R8]

Startup and running checks proposed

- Prime by the pump/package procedure. Any priming that requires opening the fluid end is an isolated, depressurized task for authorized personnel. Establish feed and wash, then start slowly with the approved low-load flow path; never start against a closed discharge.
- Verify power-end lubrication pressure promptly against the approved limit. Record inlet pressure at the TT-660 under actual pumping load, discharge pressure, pump speed, wash condition and oil-temperature trend.
- Check at startup, after load changes and at least hourly. Watch for pulsation, knocking, feed-pressure loss, overheated wash return, increasing liner leakage and rising or milky oil. Stop and investigate abnormal conditions.

TT 660 wearable life and planned care

Protect pistons liners valves and seats

Control	Crew action
Clean mud	Correct torn screens and ineffective cones promptly. Trend abrasive solids under the mud program; prevent debris entering supply tanks.
Adequate feed	Maintain tank level, unrestricted feed and charge-pump performance. Investigate cavitation early; normal sound alone does not prove adequate feed. [R7]
Wash and cooling	Keep the wash circuit clean and verify delivery to each cylinder. Correct blockage or failed wash immediately.
Correct operating envelope	Use the installed liner’s pressure/speed chart. Coordinate the required drilling flow and pressure; avoid unnecessary speed or pressure.

Control	Crew action
Early leak response	Stop for significant piston bypass or fluid-end leakage. Continuing to pump can turn an expendable repair into liner/module damage.
Fit and alignment	Mechanic verifies rod, piston and liner condition and alignment after abnormal wear. Do not keep replacing rubber without correcting the cause. [R9]
Replacement quality	Use correct matched parts and OEM assembly instructions. Inspect seats, sealing surfaces, rods and clamps; use specified tightening procedures.
Track life	Log cylinder number, part number, install/remove hours, operating conditions and failure mode. Compare hours per part and identify repeat cylinder failures.

Published lubrication reference for verification

TRI's published operating/parts excerpt, page 21, recommends API GL-5 80W-90 crankcase oil. For normal conditions it describes approximately 1,000-hour oil service, earlier for contamination, and filter replacement with each oil change or every 1,000 hours. It requires frequent level and pressure checks. TT-660 nominal capacities are 25 gal crankcase and 28 gal chainbox; the factory lube system adds 2 gal to crankcase capacity, with variations possible. Confirm applicability and chainbox lubricant separately. Fill to the specified indicator, not these numbers. [R10]

Shutdown and mechanic work

- Follow the package unloading, flushing and shutdown sequence. Flush with compatible clean fluid to the designated containment route, clean the wash circuit and protect all trapped-fluid spaces against freezing using the TRI winterization procedure.
- Mechanic handles fluid-end replacement, torque verification, power-end inspection, chain adjustment, alignment and dampener work. Dampeners can retain gas pressure after fluid pressure is relieved; only trained personnel follow the unit's procedure.
- Schedule engine, chainbox, wash-system and charge-pump services separately. A 1,000-hour crankcase reference does not establish their intervals.

Ditch Witch 120 field care

Fleet addition: Ditch Witch 120. Treat as JT120 for this section pending nameplate confirmation. If the asset is AT120, add the All Terrain inner-drive and downhole-tool service requirements from its matching manual.

Before each shift proposed HardRock routine

- Use the common rig checks on pages 7 and 8, including DAILY oil-level checks of both travel planetary/final drives and every rotary/rotation gearbox where fitted. Also check engine oil, cold coolant, hydraulic level, fuel/DEF, leaks, guards, safety controls, anchoring and pipe handling.
- Identify separate engine, hydraulic, ground-drive, rotation-drive and onboard fluid-pump lubrication systems. Record approved products and checking conditions on the asset service card.
- Inspect the compound supply, delivery nozzle and pipe joints; inspect the mud supply, swivel, hoses and wash/oiler circuits where fitted.
- Check due lubrication against the asset's chart. Keep the track-tension adjuster separate from routine greasing and use the manufacturer's stored-pressure procedure.

During drilling proposed HardRock routine

- Monitor alarms continuously. At least every two hours and at handover, check for new leaks, noise, compound-delivery failure, damaged hoses or abnormal carriage/pipe-loader operation from guarded access. Stop for unsafe conditions.

End of shift proposed HardRock routine

- Isolate before cleaning or maintenance. Clean accessible mud buildup without forcing water into electrical components, bearings or seals. Inspect pipe and wear components; perform only due, approved lubrication. Log defects and next service hours.

Model-specific lubrication charts

Vermeer 40x55 grease fitting and lubrication chart

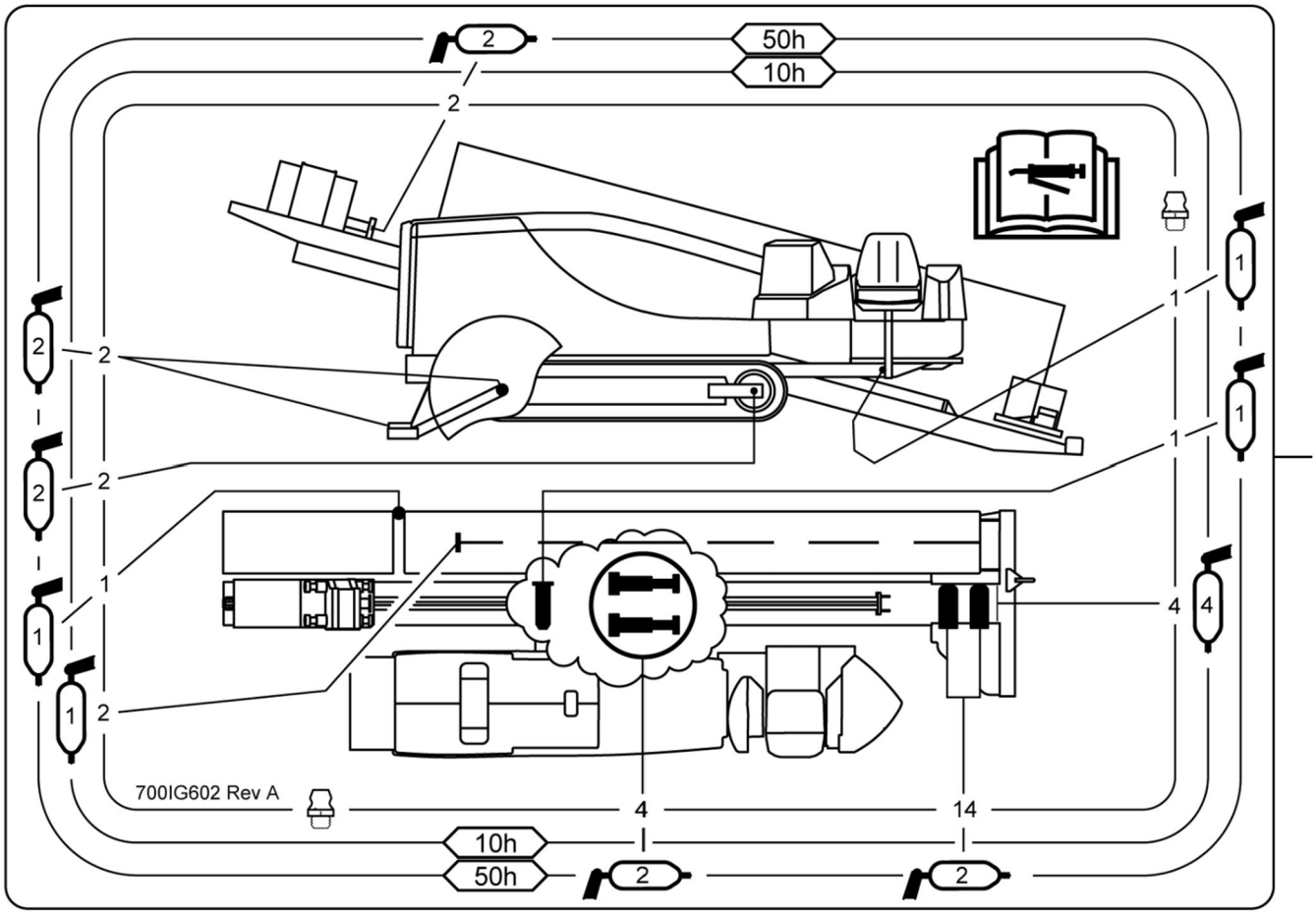


Figure: Vermeer 40x55 grease fitting locations and lubrication intervals (drawing 700IG602 Rev A). Use with the applicable Vermeer operator’s manual and the approved lubricant specifications.

Vermeer 60x90 grease fitting and lubrication chart

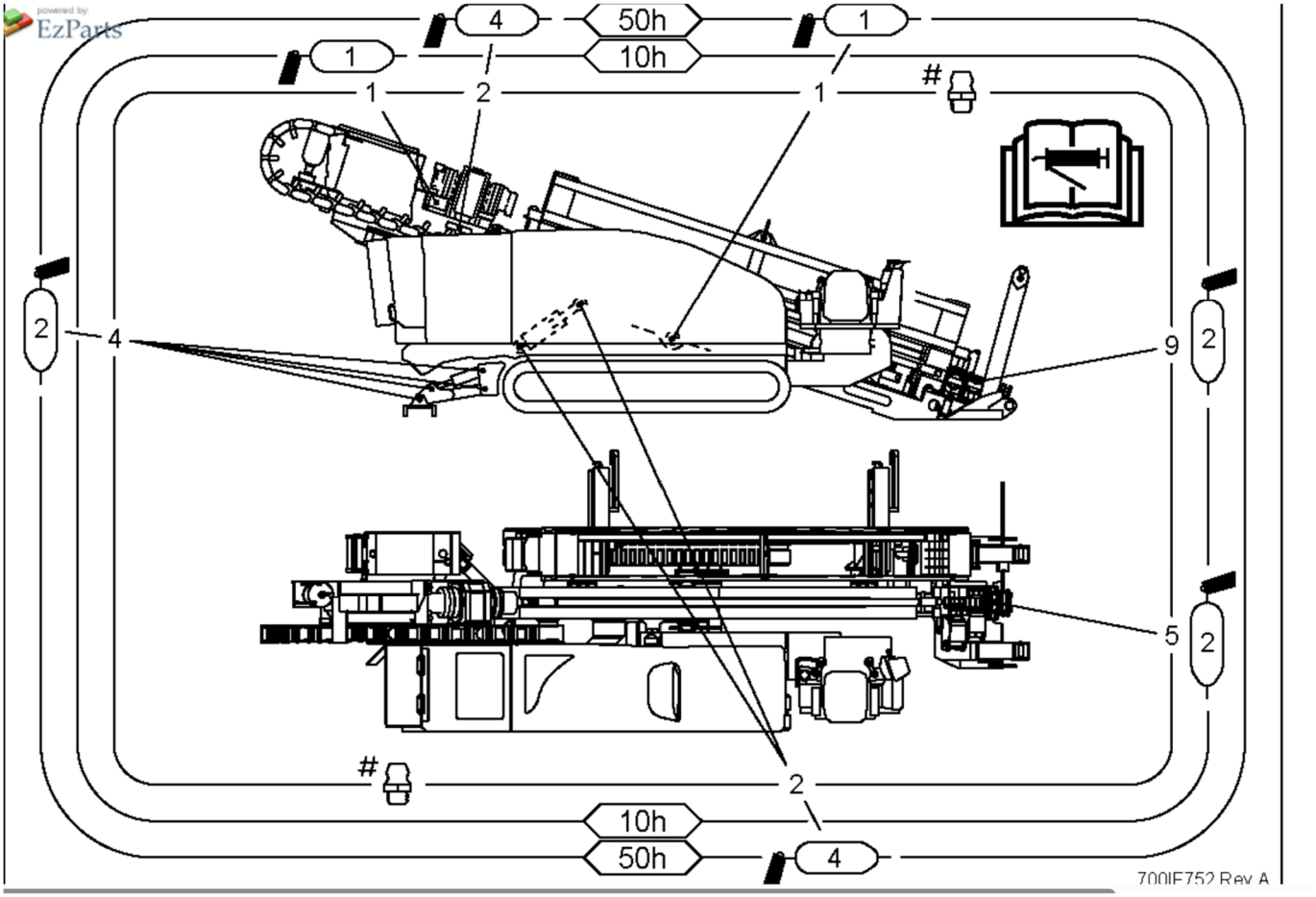


Figure: Vermeer 60x90 grease fitting locations and lubrication intervals (drawing 700IF753 Rev A). Use with the applicable Vermeer operator’s manual and the approved lubricant specifications.

Vermeer 100x140 grease fitting and lubrication chart

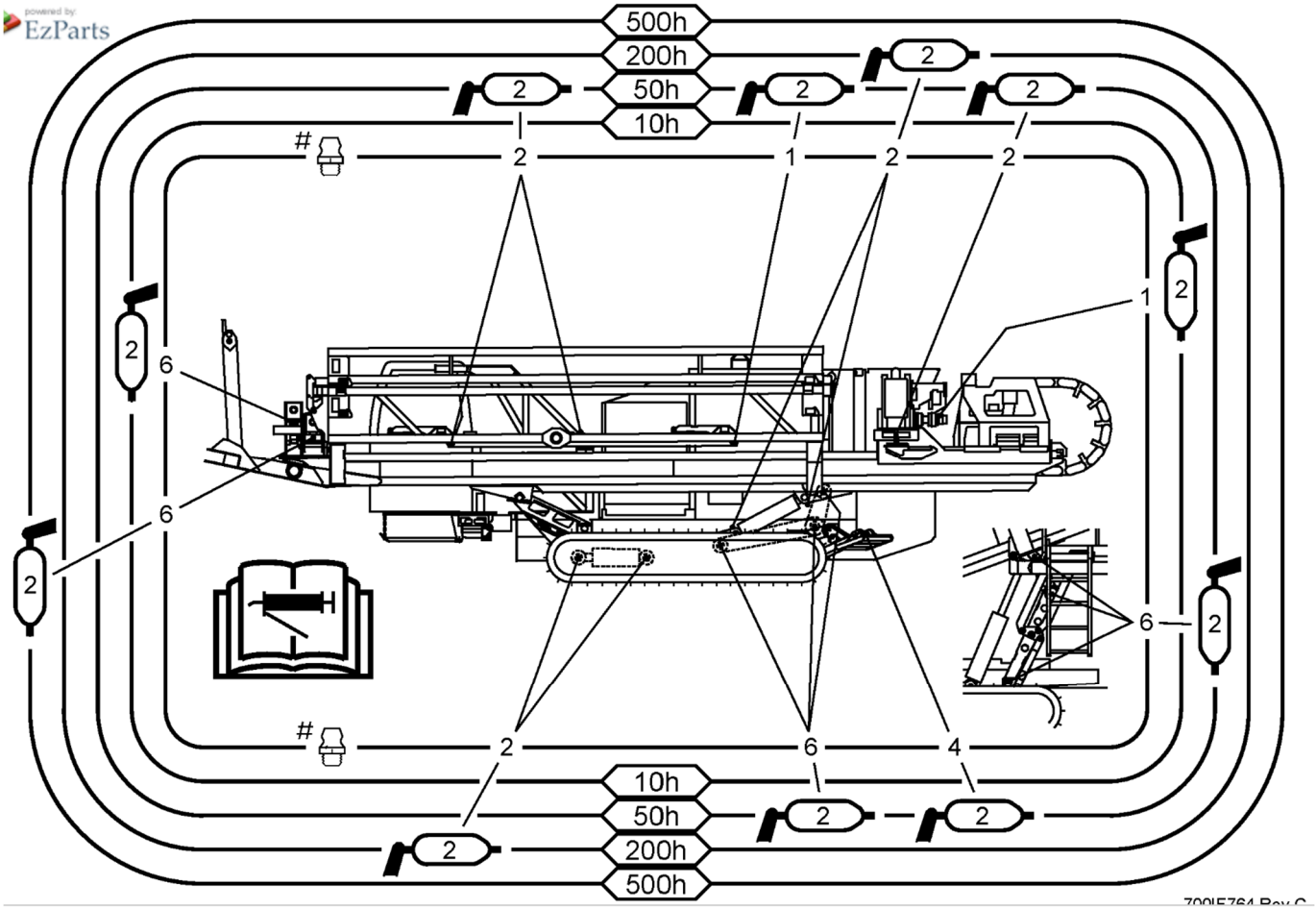


Figure: Vermeer 100x140 grease fitting locations and lubrication intervals. Use with the applicable Vermeer operator's manual and the approved lubricant specifications.

Vermeer 220x300 grease fitting and lubrication chart

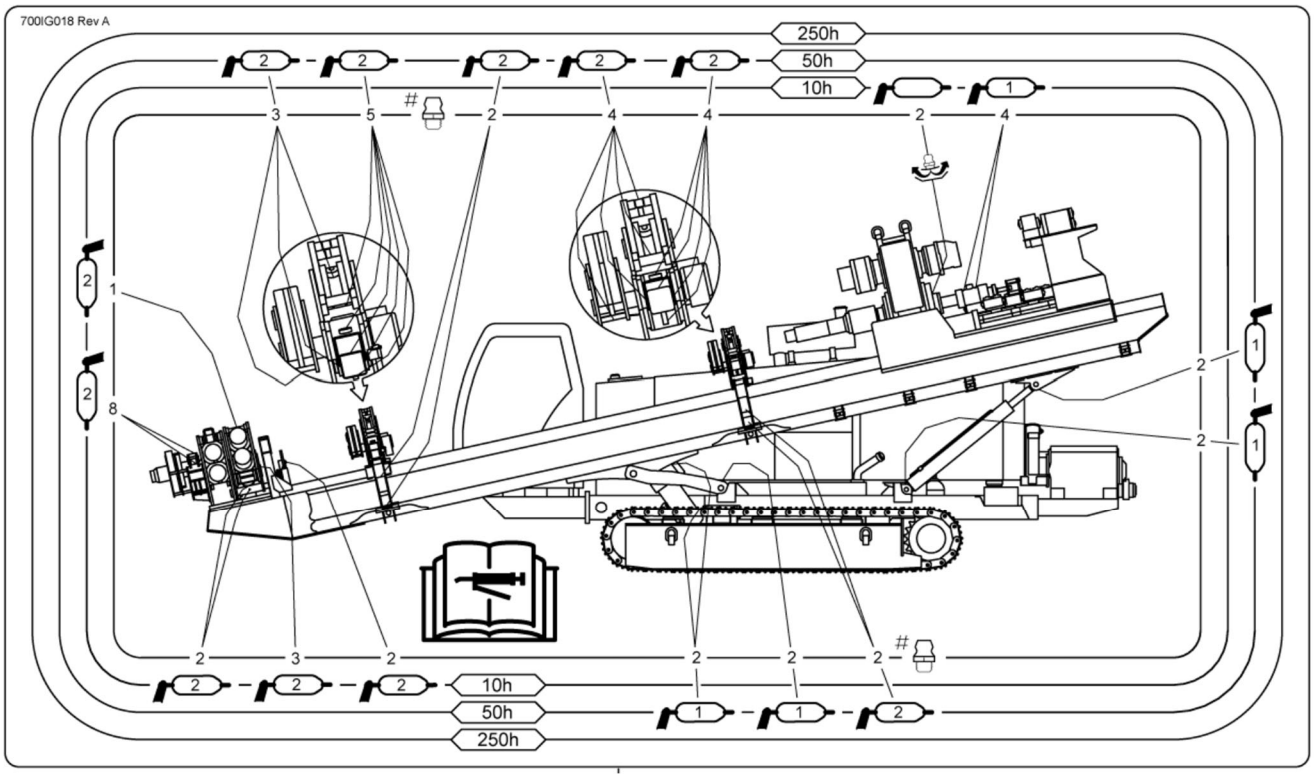


Figure: Vermeer 220x300 grease fitting locations and lubrication intervals (drawing 700IG018 Rev A). Use with the applicable Vermeer operator's manual and the approved lubricant specifications.